TEAM NOISE · CLOSED BLACK-BOX BEHAVIORAL ANALYSIS

Can AST Tell Animals From Other Sounds?

Stress-Testing Audio Spectrogram Transformer via Noise, Competing Audio & Mimicry

Investigating input–output failure modes, confidence decay, competing-sound hijacking, and vocal mimicry on **MIT/ast-finetuned-audioset-10-10-0.4593** across 5 datasets (3,659 evaluations).

PRESENTING PAIR (AI-ASSISTED)

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2:20 VIDEO ROADMAP

- **Setup:** Saturated clean benchmark vs. 4 stress tests
- **3 Visualizations:** Confidence sweep, cross-dataset view, mistake explorer
- **AI4VIS:** Collaborator & critic reflections + Phase II roadmap

AI-ASSISTED PAIR VIS4AI SETUP

Experimental Setup: Stress-Testing the Model

Why clean benchmark accuracy is insufficient to evaluate real-world audio behavior

1. CLOSED BLACK BOX

Disjoint Label Mapping

• Mapped AST's **527 AudioSet classes** to 50 ESC-50 classes.

• **Disjoint Mapping:** Excluded shared parent nodes (e.g. *"Chicken, rooster"*) so confusions come from the model, not mapping artifacts.

• Generic *"Animal"* parent node score (**0.21**) heavily underestimates specific animal detection (**0.74**).

2. BASELINE SATURATED

Clean Data Is Too Easy

• **99.0% Accuracy** on clean animal sounds (396 out of 400 correct). Only 4 errors overall.

• **Loudness Invariant:** Tested gain scaling from **-40 dB to +10 dB** across 200 clips — **100.0% accuracy**.

• High benchmark accuracy masks critical model vulnerabilities.

3. STRESS-TEST SUITE

1,659 Synthetic & Edge Clips

• **White Noise:** Gaussian static sweeps from +20 dB down to -10 dB SNR (240 clips).

• **Competing Audio:** Mixed at +10 to -10 dB with sirens, crying baby, rain, engine (1,000 clips).

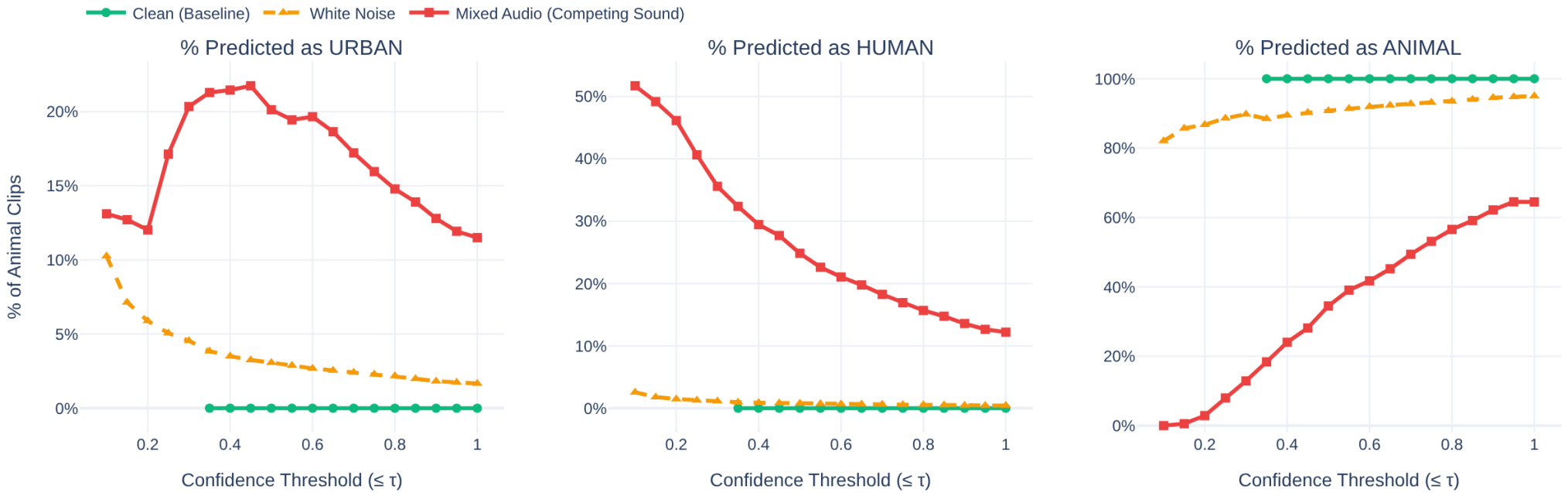
• **Vocal Mimicry:** Wild Superb Lyrebird (159 clips) & human vocal imitations (20 clips).

VIS 1 CONFIDENCE SWEEP

Finding 1: Static Noise vs. Competing Sounds

Threshold sweep (τ ≤ threshold) exposes fundamentally different prediction drift modes

Cumulative Confidence Sweep: As Animal Confidence Drops, Predictions Drift into Urban & Human Categories



- **Clean Baseline (Green)**

Perfect **100% animal predictions** across all confidence levels. Never confuses clean animals with urban or human sounds.

- **White Noise (Yellow)**

Noise lowers confidence (mean **0.11** at -10 dB), yet **82% to 95% of predictions stay in the Animal family**. Static rarely causes category drift.

- **Mixed Audio (Red)**

Severe hijacking: low-confidence animal clips drift to **>22% Urban (siren)** and **~52% Human (crying baby)** as distractors dominate.

Plot: `viz_plots/cumulative_confidence_sweep_urban_human_animal.png`

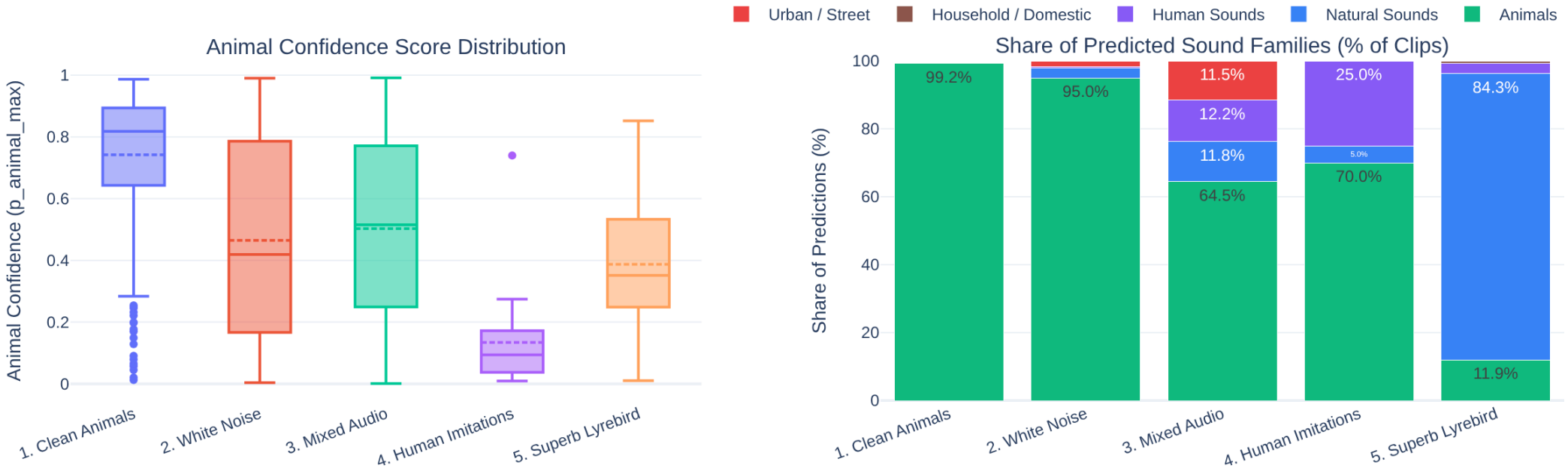
VIS 2 CROSS-DATASET COMPARISON

⏱️ 1:00 – 1:25 (25s) · Slide 4/6

Finding 2: Confidence vs. Sound Family Predictions

Pairing confidence distributions with predicted sound families reveals hidden vulnerability

Cross-Dataset Comparison: Clean Animals vs. Perturbed Audio vs. Imitations & Lyrebird



- **Noise vs. Mixed Audio**

White noise maintains **95.0% Animal predictions** despite broad confidence drops. In mixed audio, animal share falls to **64.5%**.

- **Human Vocal Imitations**

Looks like 70% Animal predictions, but **confidence collapsed to median ~0.10**. 25% are detected as Human (whistling, humming).

- **Superb Lyrebird Mimicry**

84.3% Natural bird song. 11.9% Animal predictions reflect real mimicry: AST correctly detects frog in clip XC738172.

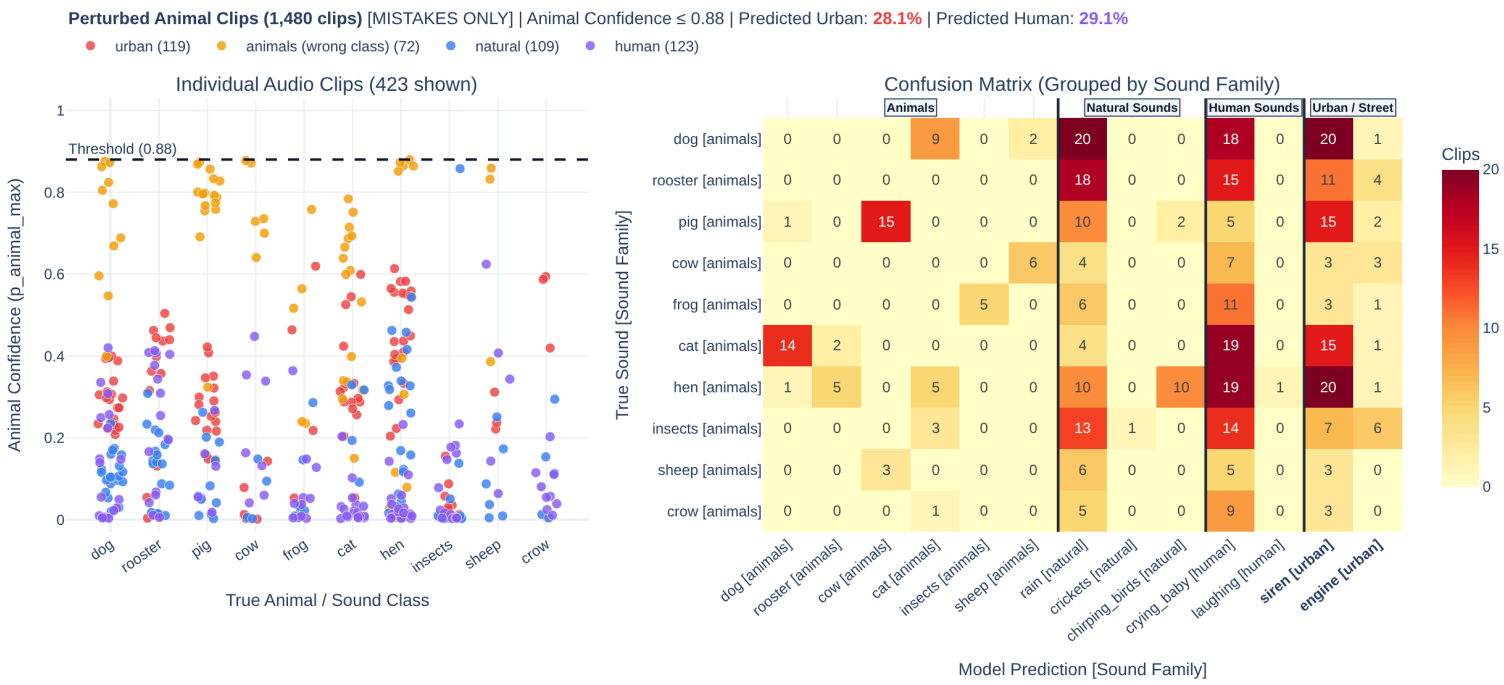
Plot: `viz_plots/cross_dataset_comparison_confidence_and_sound_families.png`

VIS 3 MISTAKE EXPLORER

⏱️ 1:25 – 1:55 (30s) · Slide 5/6

Finding 3: Distractor Hijacking & Hen Vulnerability

Mistakes-only matrix (τ ≤ 0.88): 443 of 455 mixing errors are simply the distractor winning



- **97.4% Distractor Takeovers**

Out of 455 mixing errors, **443 simply name the mixed-in sound** (crying baby, siren, rain). Only 12 are genuine 3rd-class confusions.

- **Engine Is Ignored**

Unlike siren and crying baby which overpower the animal, **engine is ignored 91% of the time** — AST continues detecting the animal sound.

- **Hen Loses 20 / 20 Mixes**

Severe asymmetry: **hen loses to rooster in 20 out of 20 mixes**, even when hen is **+10 dB louder**! Rooster survives ~50% of mixes.

Plot: `viz_plots/confusion_matrix_perturbed_animals_conf_le_0.88_mistakes_only.png`

AI4VIS & SYNTHESIS TRANSITION TO PHASE II

AI4VIS Reflection & Looking Ahead to Phase II

Lessons from human–AI collaboration and opening the black box in Phase II

AI4VIS INTERACTION

AI as Collaborator & Critic

• **Adversarial Data Auditor:** Prompted LLM to falsify our conclusions against raw prediction CSVs. AI caught that 443/455 errors were just the distractor winning.

• **Refining AI Ideas:** When AI suggested dropping weak predictions via a static cutoff, we turned it into a **continuous threshold sweep** ($\tau \leq$ threshold).

• **Comprehension Debt:** Rapid code generation required manual auditing and **listening to raw audio clips** to verify unexpected findings.

SYNTHESIS & NEXT STEPS

Pair Convergence & Phase II

• **Independent Convergence:** Both Human and AI pairs identified **hen** as AST's primary vulnerability (responsible for 2 of 4 clean errors; loses 20/20 mixes to rooster).

• **Phase II (Opening Black Box):** Use **Attention Rollout & patch attributions** to understand why sirens and crying babies overpower animal attention heads.

• **Role Swap:** Gurmehak & Elif take the AI-assisted pipeline; Asiya & Rachid switch to human-only exploration.

Thank You!